

# Jaejun Lee (이재준, 李在竣)

PHD STUDENT · SCHOOL OF COMPUTING, KAIST

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## Education

### KAIST

PH.D., SCHOOL OF COMPUTING

Daejeon

2023.02 - present

- Big Data Intelligence Lab (KAIST BDI Lab)
- Advisor: Professor Joyce Jiyoung Whang

### KAIST

M.S., SCHOOL OF COMPUTING

Daejeon

2021.08 - 2023.02

- Big Data Intelligence Lab (KAIST BDI Lab)
- Advisor: Professor Joyce Jiyoung Whang

### KAIST

B.S., SCHOOL OF COMPUTING

Daejeon

2017.02 - 2021.08

- Double Major: Department of Mathematical Sciences

## Publications

† equal contribution, \* corresponding author

1. **Jaejun Lee** and Joyce Jiyoung Whang\*, “Learn When and Where to Connect: Adaptive Virtual Nodes for Dynamic Message Passing on Graphs”, *Proceedings of the 32nd SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. To be published.
2. **Jaejun Lee**, Seheon Kim, and Joyce Jiyoung Whang\*, “Generative Representation Learning on Hyper-relational Knowledge Graphs via Masked Discrete Diffusion”, *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. To be published.
3. **Jaejun Lee** and Joyce Jiyoung Whang\*, “Structure Is All You Need: Structural Representation Learning on Hyper-Relational Knowledge Graphs”, *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, pages 33765-33783, July 2025.
4. Minsung Hwang, **Jaejun Lee**, and Joyce Jiyoung Whang\*, “Stability and Generalization Capability of Subgraph Reasoning Models for Inductive Knowledge Graph Completion”, *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, pages 26376-26411, July 2025.
5. **Jaejun Lee**, Minsung Hwang, and Joyce Jiyoung Whang\*, “PAC-Bayesian Generalization Bounds for Knowledge Graph Representation Learning”, *Proceedings of the 41st International Conference on Machine Learning (ICML)*, pages 26589-26620, July 2024.
6. **Jaejun Lee**, Chanyoung Chung, Hochang Lee, Sungho Jo, and Joyce Jiyoung Whang\*, “VISTA: Visual-Textual Knowledge Graph Representation Learning”, *Findings of the Association for Computational Linguistics: EMNLP 2023 (Findings of EMNLP)*, pages 7314-7328, December 2023.
7. Chanyoung Chung†, **Jaejun Lee**†, and Joyce Jiyoung Whang\*, “Representation Learning on Hyper-Relational and Numeric Knowledge Graphs with Transformers”, *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, pages 310-322, August 2023.
8. **Jaejun Lee**, Chanyoung Chung, and Joyce Jiyoung Whang\*, “InGram: Inductive Knowledge Graph Embedding via Relation Graphs”, *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pages 18796-18809, July 2023.
9. Ji Ho Kwak†, **Jaejun Lee**†, Sungho Jo\*, and Joyce Jiyoung Whang\*, “Semantic Grasping via a Knowledge Graph of Robotic Manipulation: A Graph Representation Learning Approach”, *IEEE Robotics and*

Teaching Experience \_\_\_\_\_

TUTORIALS

- Assistant Tutor of “Knowledge Discovery with Knowledge Graphs: From Structural Embeddings to Generative Reasoning(KnowKG)”, To Appear in *SIGKDD Conference on Knowledge Discovery and Data Mining(KDD)*, Tutorial, August 2026.
- Assistant Tutor of “Key Facets in Modern Knowledge Graph Representation Learning (KeyKGRL)”, *International Semantic Web Conference (ISWC)*, Tutorial, November 2025.

TEACHING ASSISTANT

|                                                                             |                                      |       |
|-----------------------------------------------------------------------------|--------------------------------------|-------|
| CS471 Graph Machine Learning and Mining                                     | Spring 2023, 2024 and 2025           | KAIST |
| CS376 Machine Learning                                                      | Fall 2021, 2022, 2023, 2024 and 2025 | KAIST |
| CS665(DS532) Advanced Data Mining                                           | Fall 2023                            | KAIST |
| CS492 Special Topics in Computer Science: Graph Machine Learning and Mining | Spring 2022                          | KAIST |

Professional Services \_\_\_\_\_

PEER REVIEW

Conference Reviewer: NeurIPS, ICML (Gold Reviewer 2026), KDD, ACL ARR

Journal Reviewer: TKDD, TMLR